

Communication-Efficient and Robust Multi-Robot Mapping Framework

(Abstract) Multi-robot mapping across multiple robots accelerates exploration and improves coverage efficiency by maintaining global consistency. However, multi-robot mapping requires exchanging massive sensor data between robots, making communication the bottleneck, especially in communication-constrained environments. Therefore, we present a hybrid framework combining centralized coordination with distributed selective data exchange that reduces inter-robot communication by up to 1,000 $\times$ while maintaining alignment accuracy. Our key insight is that only a small subset of carefully selected scans is sufficient for robust multi-robot mapping. We formulate this as solving a three-stage cascade problem on an exchange graph, where vertices represent robot keyframes and edges denote candidate inter-robot loops. Through three cascade stages, we select a sequentially overlapped, minimal-transmission-cost, and mutually consistent optimal scan subset that preserves alignment quality while reducing communication. Unlike existing approaches that transmit whole scans, which require GB-scale data exchange, or employ naïve downsampling, our approach exchanges only MB-scale data while achieving comparable alignment accuracy. The evaluation shows up to 99.98 % reduction in data exchange (e.g., from 7.0 GB to 1.7 MB in the large-scale HeLiPR dataset), while maintaining real-time performance.

1. Introduction

Multi-robot systems achieve remarkable collective intelligence by seamlessly merging their individual maps into unified environmental representations. Similar to solving a puzzle, this requires aligning each robot's map piece and merging them to form a unified global map. Light detection and ranging (LiDAR)-based multi-robot mapping addresses this by estimating alignments through the association of metadata (i.e., global descriptors) and full data (i.e., scans) of each robot's map piece. A central entity, known as a server (or central robot), typically coordinates these frameworks, managing not only global descriptors matching and loop closure detection but also allocating shared resources such as bandwidth, battery power, memory, and computational capacity. Such frameworks are essential for large-scale missions where maintaining a consistent global frame boosts exploration efficiency.

However, one major challenge still hinders practical deployment: servers must exchange massive amounts of data, exhausting bandwidth and battery. To address these issues, existing approaches follow two main strategies: (i) centralized or (ii) distributed frameworks. The centralized frameworks aggregate all robot data on a server for global optimization, providing effective coordination. However, they require continuous high-bandwidth communication and create scalability bottlenecks as data volume grows. The distributed frameworks reduce server reliance through peer-to-peer data exchange during robot encounters, offering better scalability. However, they often transmit entire data backlogs at each encounter, creating a bursty communications load.

Although these frameworks adopt resource-saving techniques, such as downsampling or keyframing, to mitigate these problems, they often sacrifice geometric detail or discard informative loops, thereby degrading map quality. Moreover, most works assume stable communication links, overlooking bandwidth constraints common in real-world operations. This creates a gap between experimental demonstrations and field-deployable systems.

To bridge this gap, we propose a LiDAR-based multi-robot

mapping specifically designed for bandwidth-constrained environments. Unlike existing approaches that strictly follow either a centralized or a distributed paradigm, our proposed method blends the advantages of centralized and distributed approaches by employing a centralized server for global coordination while using selective and bandwidth-efficient data sharing to reduce communication load.

The main contributions of this work are as follows. First, we propose a multi-robot mapping that achieves high mapping accuracy with lower communication and computation costs, making it practical for communication-constrained field deployments. Second, we formulate a communication-efficient exchange graph to select an optimal scan subset and suppress redundant scan transfers.

2. Communication-Efficient and Robust Multi-Robot Mapping Framework

The entire process of our proposed framework proceeds in six stages: (i) intra-robot SLAM, (ii) inter-robot place recognition, (iii) inter-robot sequential matching, (iv) balanced minimum vertex cover in the exchange graph, (v) maximum edge-weighted clique selection in the exchange graph, and (vi) submap-based map merging. From intra-robot SLAM at the local robot-level to multi-robot mapping at the server-level, each robot and the server communicate organically and refine candidate loops incrementally, ensuring that only the most informative and mutually consistent data are exchanged.

First, each robot runs intra-robot SLAM to generate descriptors, poses, and scans, producing local maps (Fig. 1.(a)). Second, the server receives descriptors and poses via communication and performs descriptor-based similarity checking to assess the strength of place matches between robots (Fig. 1.(b)). Based on the inter-robot place recognition loop set, the server forms an exchange graph and initiates a three-stage cascade optimization problem (i.e., step (A), step (B), and step (C)). In step (A), the server introduces sequential matching for robust inter-robot place recognition by detecting temporally contiguous correspondence (Fig. 1.(c)). In

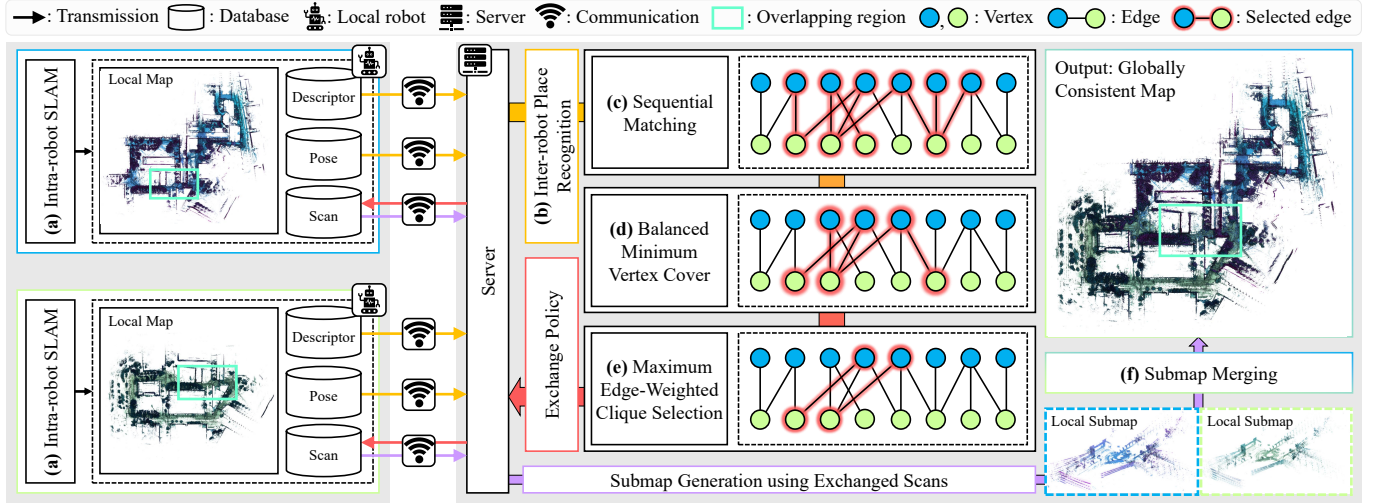


Fig. 1. Overall pipeline of our proposed communication-efficient and robust multi-robot mapping.

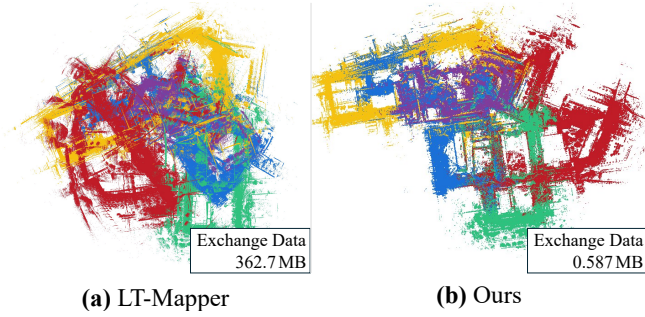


Fig. 2. Five robot mapping comparison between LT-Mapper [2] and our method on the Kimera-Multi dataset [3] with data exchange.

step (B), the server applies balanced minimum vertex cover to the sequentially matched scans to select scans that maintain minimal overall communication cost while ensuring balanced cost distribution between robots (Fig. 1.(d)). In step (C), the server selects the final optimal scans from the minimal cost scans by maintaining geometrically and perceptually consistent correspondences (Fig. 1.(e)). Subsequently, the server creates a policy instructing the robots to transmit only these optimal scans, and the robots transmit partial scans according to this policy instead of whole scans. Finally, the server constructs submaps from these partial scans and generates a globally consistent map through final submap merging (Fig. 1.(f)).

Fig. 2 and Fig. 3 shows robust multi-robot map merging, reducing data exchange while maintaining mapping quality. These results validate that our method achieves communication reduction while preserving global consistency.

3. Conclusion

We present a communication-efficient multi-robot mapping that reduces data exchange by up to 99.98% while maintaining mapping accuracy. Our novel exchange graph formulation enables systematic identification of minimal scan subsets through three-stage cascade optimization, achieving robust map merging with only MB-scale communication instead of GB-scale data exchange. The framework successfully bridges the gap between experimental demonstrations

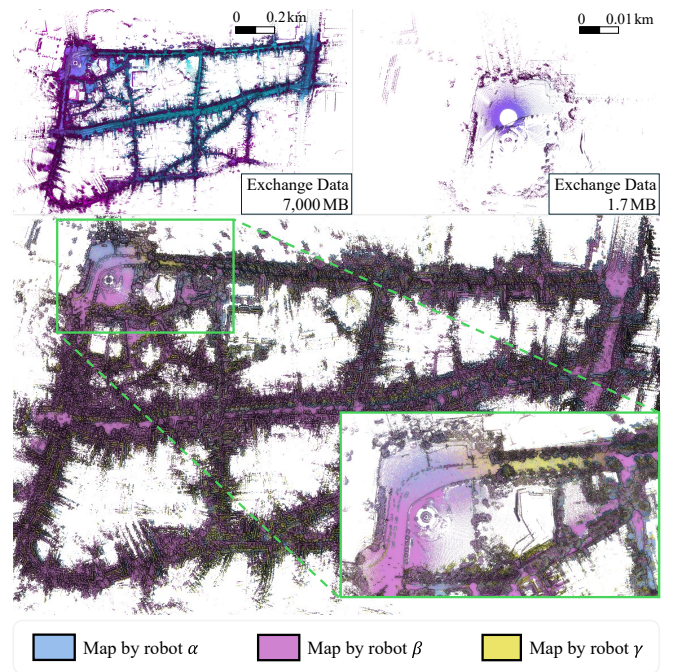


Fig. 3. Multi-robot mapping comparison on HeLiPR dataset [1]. Top: robot α 's maps using LT-Mapper (left, 7,000 MB) [2] vs. our approach (right, 1.7 MB). Bottom: final merged map showing robust integration of three robot contributions without visual distortions.

and real-world deployments in bandwidth-limited environments or large-scale missions.

REFERENCES

- [1] M. Jung, W. Yang, D. Lee, H. Gil, G. Kim, and A. Kim. Helipr: Heterogeneous lidar dataset for inter-lidar place recognition under spatiotemporal variations. *The International Journal of Robotics Research*, 43(12):1867–1883, 2024.
- [2] G. Kim and A. Kim. Lt-mapper: A modular framework for lidar-based lifelong mapping. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 7995–8002. IEEE, 2022.
- [3] Y. Tian, Y. Chang, L. Quang, A. Schang, C. Nieto-Granda, J. P. How, and L. Carlone. Resilient and distributed multi-robot visual slam: Datasets, experiments, and lessons learned. 2023.